



Customer Churn Analysis – SQL Project

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Introduction

Customer churn directly affects revenue and growth. The business wants to understand which customers churn, why they churn, and how to prevent it using customer behavior, engagement, pricing, and support data.



Objective

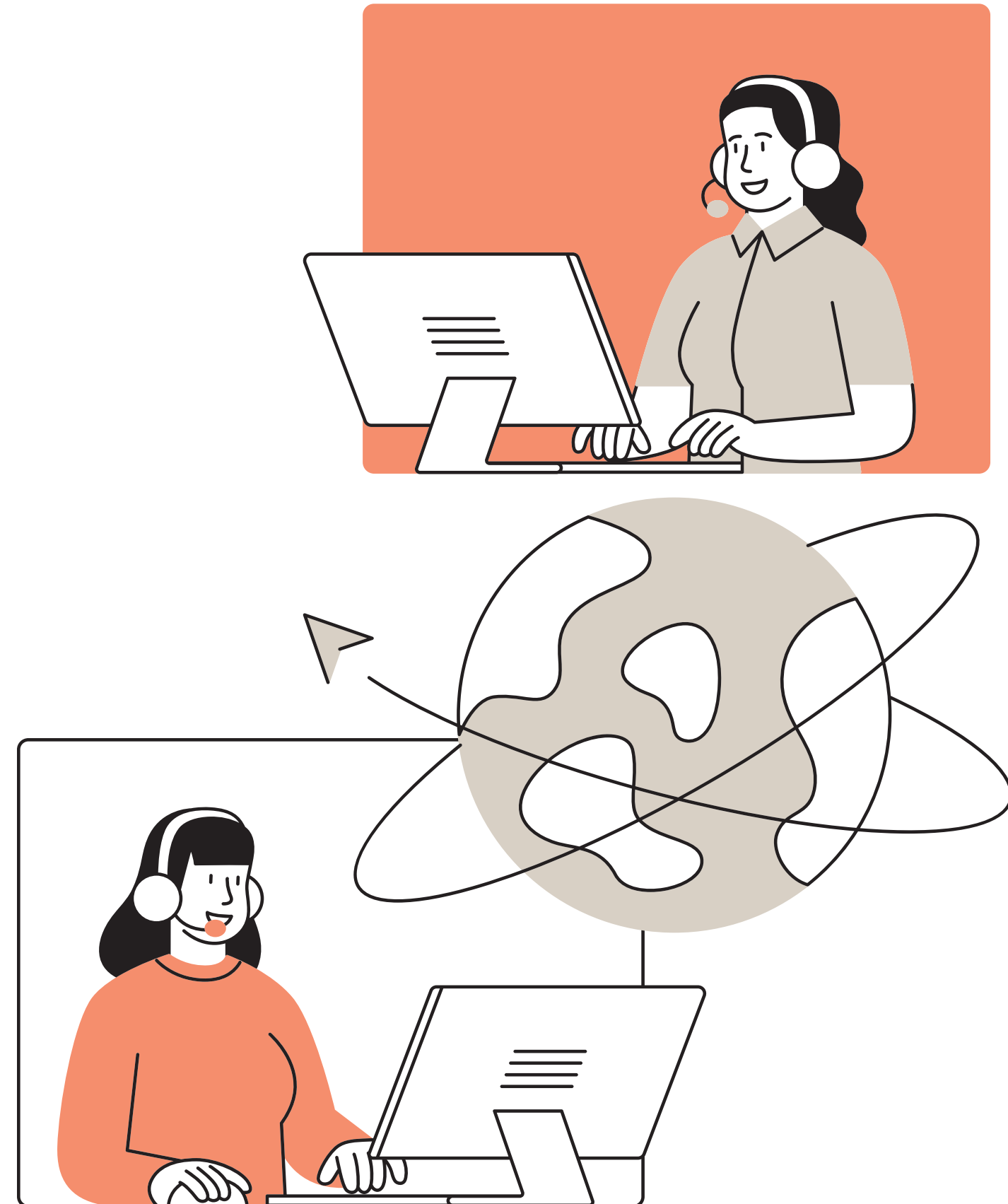
Measure churn rate

Identify churn drivers

Compare churned vs retained customers

Detect high-risk customers early

Provide actionable business insights



What is the overall churn rate?

```
1 • SELECT
2     COUNT(*) AS total_customers,
3     SUM(churn) AS churned_customers,
4     ROUND(SUM(churn)*100.0/COUNT(*),2) AS churn_rate_percent
5 FROM cc.customer_churn;
```

Insight

- Establishes the baseline health of the business
- Used as a primary KPI
- Even small churn % can cause major revenue loss



	total_customers	churned_customers	churn_rate_percent
▶	10000	1021	10.21

Churn distribution by gender

```
1 • SELECT gender,  
2     ROUND(SUM(churn)*100.0/COUNT(*),2) AS churn_rate  
3     FROM cc.customer_churn  
4     GROUP BY gender;
```

Insight

- Churn is not strongly gender-driven
- Indicates churn is influenced more by behavior and experience

	gender	churn_rate
▶	Male	10.31
	Female	10.11



Which customer segment has the highest churn?



```
1 • SELECT customer_segment,  
2       COUNT(*) AS customers,  
3       SUM(churn) AS churned,  
4       ROUND(SUM(churn)*100.0/COUNT(*),2) AS churn_rate  
5 FROM cc.customer_churn  
6 GROUP BY customer_segment;
```

Insight

- Individual customers churn more than SME
- SMEs usually show higher dependency and loyalty
- Retention efforts should prioritize Individual users

	customer_segment	customers	churned	churn_rate
▶	SME	3029	331	10.93
	Individual	5984	597	9.98
	Enterprise	987	93	9.42

Does contract type affect churn?

```
1 • SELECT contract_type,  
2     ROUND(SUM(churn)*100.0/COUNT(*),2) AS churn_rate  
3 FROM cc.customer_churn  
4 GROUP BY contract_type;
```



Insight

- Monthly contracts show higher churn
- Long-term contracts reduce churn risk

	contract_type	churn_rate
▶	Monthly	10.33
	Yearly	10.34
	Quarterly	9.93

How does tenure impact churn?

```
SELECT churn,  
       AVG(tenure_months) AS avg_tenure  
FROM cc.customer_churn  
GROUP BY churn;
```



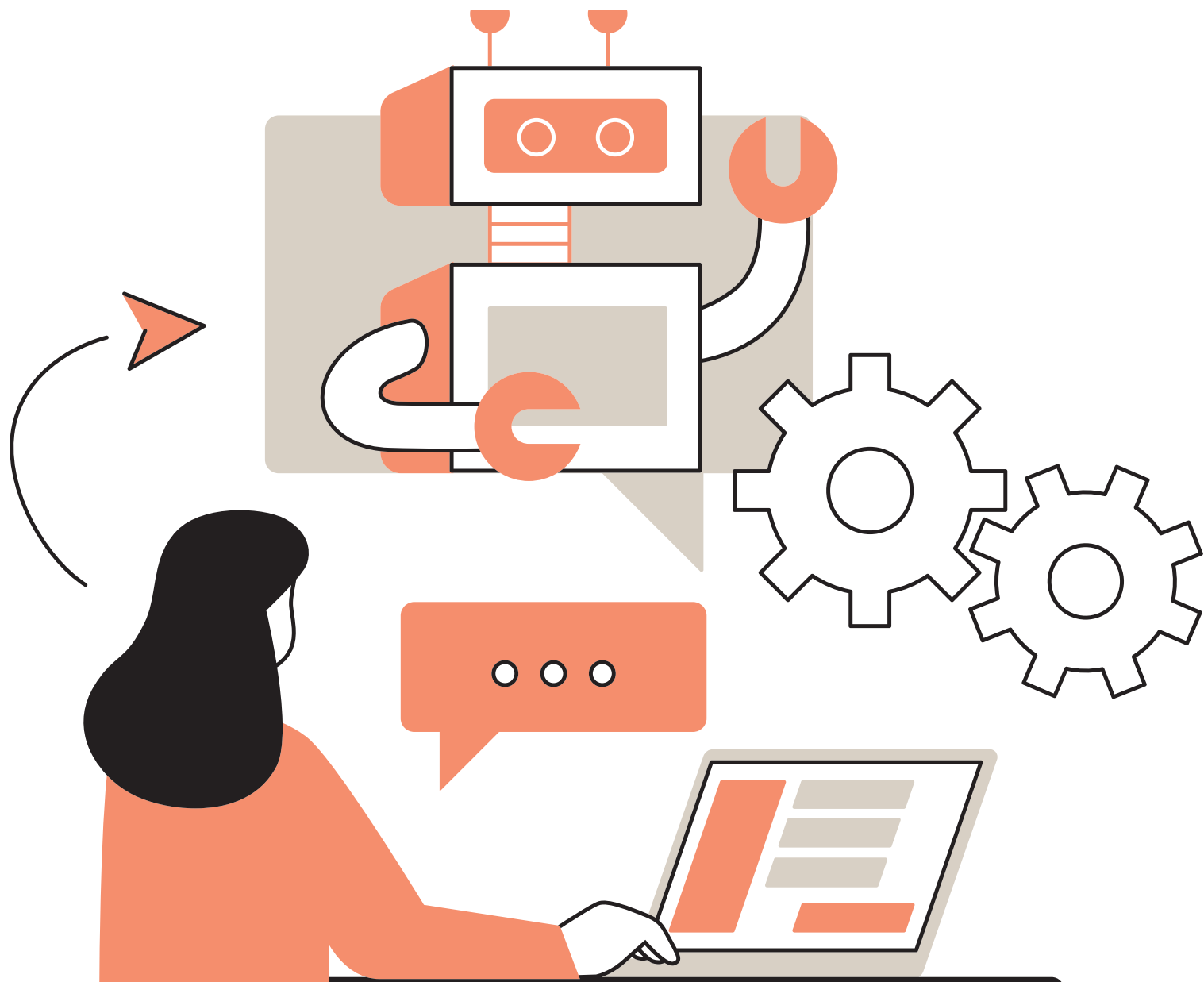
Insight

- Churned users have lower tenure
- Early lifecycle customers are high-risk
- Onboarding experience is critical

	churn	avg_tenure
▶	0	30.8304
	1	24.2223

Does engagement differ for churned customers?

```
• SELECT churn,
    AVG(monthly_logins) AS avg_logins,
    AVG(weekly_active_days) AS avg_active_days,
    AVG(avg_session_time) AS avg_session_time
FROM cc.customer_churn
GROUP BY churn;
```



Insight

- Churned users show lower engagement
- Engagement metrics are strong churn indicators

	churn	avg_logins	avg_active_days	avg_session_time
▶	0	19.9982	3.4862	15.198621713286247
	1	16.8031	3.4192	15.09350823488638

Does inactivity lead to churn?

```
1 • SELECT churn,  
2     AVG(last_login_days_ago) AS avg_days_inactive  
3 FROM cc.customer_churn  
4 GROUP BY churn;
```

Insight

- Churned customers stay inactive longer
- Login gaps signal declining interest

	churn	avg_days_inactive
▶	0	9.3821
	1	10.5877



Do discounts reduce churn?

```
1 • SELECT discount_applied,  
2     ROUND(SUM(churn)*100.0/COUNT(*),2) AS churn_rate  
3 FROM cc.customer_churn  
4 GROUP BY discount_applied;
```



Insight

- Discounted customers churn less
- Discounts help retain price-sensitive users

	discount_applied	churn_rate
▶	Yes	9.61
	No	10.47

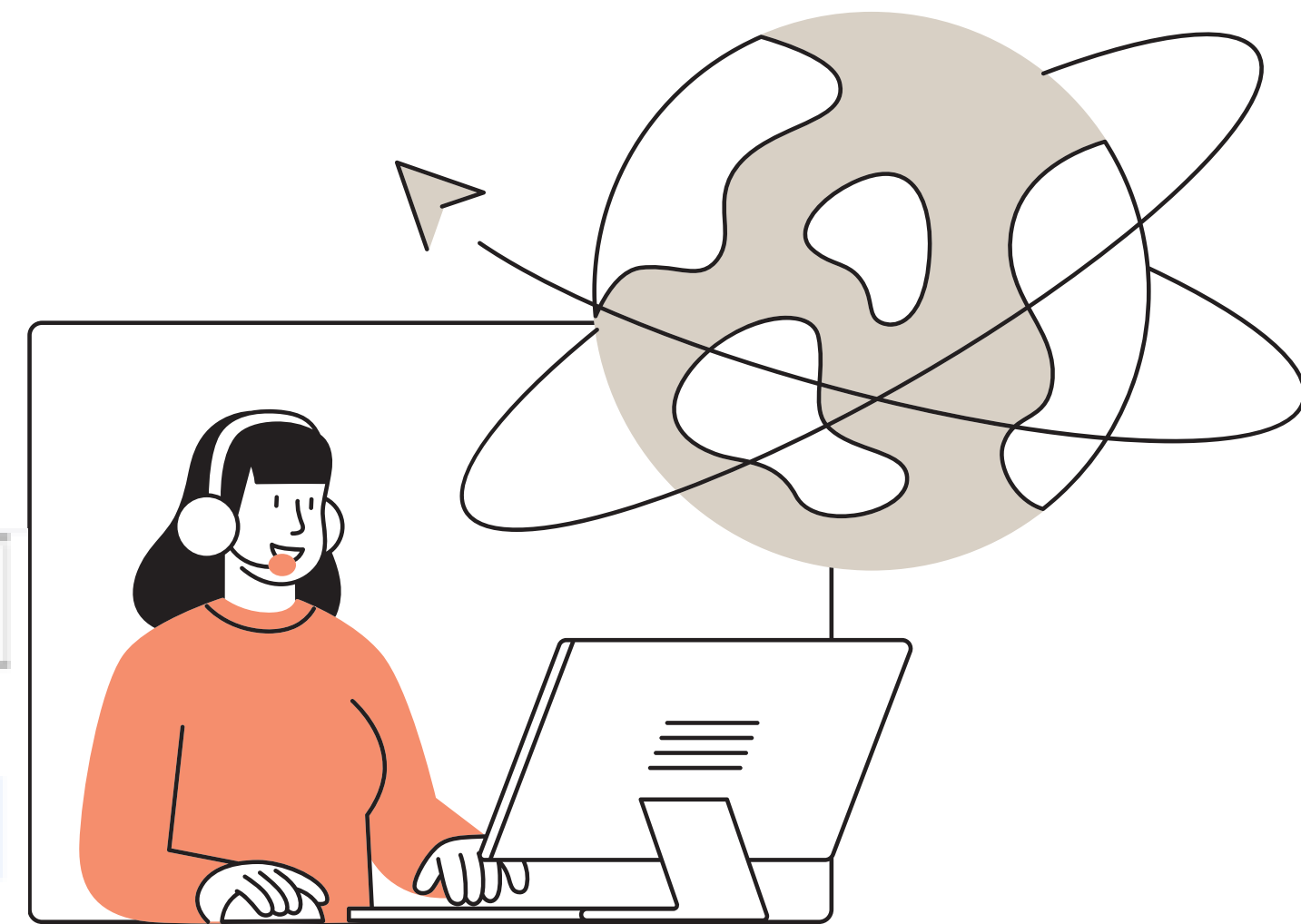
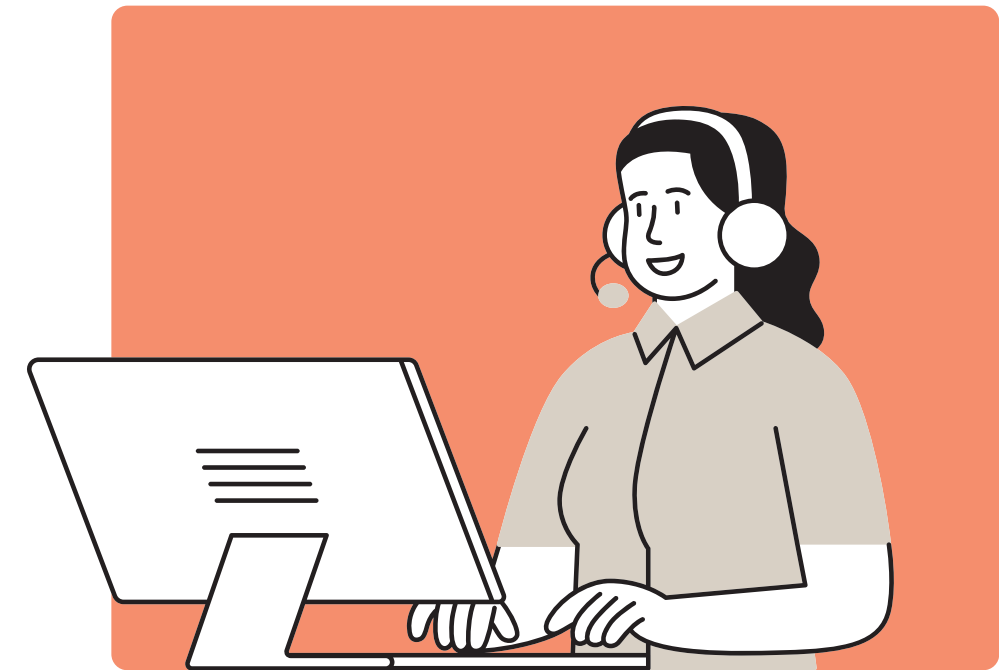
Effect of recent price increases?

```
SELECT price_increase_last_3m,  
       ROUND(SUM(churn)*100.0/COUNT(*),2) AS churn_rate  
FROM cc.customer_churn  
GROUP BY price_increase_last_3m;
```

Insight

- Recent price increases increase churn probability
- Communication before price hikes is important

	price_increase_last_3m	churn_rate
▶	No	10.17
	Yes	10.39



Which complaint types drive churn?

```
1 • SELECT complaint_type,  
2     ROUND(SUM(churn)*100.0/COUNT(*),2) AS churn_rate  
3 FROM cc.customer_churn  
4 GROUP BY complaint_type;
```

Insight

- Billing and Technical issues are major churn drivers
- These need priority resolution



	complaint_type	churn_rate
▶	Service	9.95
	Billing	10.18
	Technical	10.32
	None	10.32

Engagement Score using weighted metrics

```
1 • SELECT customer_id,  
2     (monthly_logins * 0.4 +  
3     weekly_active_days * 0.3 +  
4     features_used * 0.3) AS engagement_score  
5 FROM cc.customer_churn;
```

Insight

- Converts multiple engagement metrics into a single score
- Simplifies churn analysis



	customer_id	engagement_score
▶	CUST_00001	14.0
	CUST_00002	4.6
	CUST_00003	10.9
	CUST_00004	16.3
	CUST_00005	10.4
	CUST_00006	1.9
	CUST_00007	11.4
	CUST_00008	11.3
	CUST_00009	14.6
	CUST_00010	9.6
	CUST_00011	16.3
	CUST_00012	12.0
	CUST_00013	13.9
	CUST_00014	10.6
	CUST_00015	17.6
	CUST_00016	7.4
	CUST_00017	4.6
	CUST_00018	5.7

Engagement score vs churn (CTE)



```
1  ● ○ WITH engagement AS (  
2      SELECT churn,  
3          (monthly_logins * 0.4 +  
4            weekly_active_days * 0.3 +  
5            features_used * 0.3) AS engagement_score  
6      FROM cc.customer_churn  
7  )  
8  SELECT churn,  
9      AVG(engagement_score) AS avg_engagement  
10 FROM engagement  
11 GROUP BY churn;
```

	churn	avg_engagement
▶	0	10.54118
	1	9.25495

Insight

- Churned users have significantly lower engagement scores

Customer churn risk scoring model

```
1 • SELECT customer_id,  
2      (  
3          CASE WHEN monthly_logins < 5 THEN 2 ELSE 0 END +  
4          CASE WHEN usage_growth_rate < 0 THEN 2 ELSE 0 END +  
5          CASE WHEN csat_score <= 3 THEN 2 ELSE 0 END +  
6          CASE WHEN last_login_days_ago > 15 THEN 2 ELSE 0 END +  
7          CASE WHEN support_tickets > 2 THEN 1 ELSE 0 END  
8      ) AS churn_risk_score  
9 FROM cc.customer_churn;
```

	customer_id	churn_risk_score
▶	CUST_00001	1
	CUST_00002	4
	CUST_00003	4
	CUST_00004	5
	CUST_00005	2
	CUST_00006	6
	CUST_00007	0
	CUST_00008	0
	CUST_00009	4
	CUST_00010	0
	CUST_00011	2
	CUST_00012	0
	CUST_00013	2
	CUST_00014	2
	CUST_00015	2
	CUST_00016	2

Insight

- Converts behavioral signals into a numeric churn risk
- Very useful for proactive retention



Identify high-risk active customers

```
1  WITH risk AS (  
2      SELECT customer_id, churn,  
3          (  
4              CASE WHEN monthly_logins < 5 THEN 2 ELSE 0 END +  
5              CASE WHEN usage_growth_rate < 0 THEN 2 ELSE 0 END +  
6              CASE WHEN csat_score <= 3 THEN 2 ELSE 0 END +  
7              CASE WHEN last_login_days_ago > 15 THEN 2 ELSE 0 END +  
8              CASE WHEN support_tickets > 2 THEN 1 ELSE 0 END  
9          ) AS risk_score  
10     FROM cc.customer_churn  
11 )  
12 SELECT *  
13 FROM risk  
14 WHERE churn = 0 AND risk_score >= 6;
```

	customer_id	churn	risk_score
▶	CUST_00006	0	6
	CUST_00026	0	7
	CUST_00028	0	6
	CUST_00032	0	6
	CUST_00044	0	6
	CUST_00073	0	6
	CUST_00076	0	6
	CUST_00087	0	6
	CUST_00102	0	6
	CUST_00125	0	7
	CUST_00150	0	6
	CUST_00206	0	6
	CUST_00237	0	6
	CUST_00267	0	6
	CUST_00281	0	6
	CUST_00289	0	6

Insight

- Identifies customers likely to churn soon
- Ideal targets for retention campaigns



Revenue-based customer segmentation (NTILE)

```
1 • SELECT customer_id,  
2         total_revenue,  
3         NTILE(4) OVER (ORDER BY total_revenue DESC) AS revenue_quartile  
4 FROM cc.customer_churn;
```

	customer_id	total_revenue	revenue_quartile
►	CUST_07827	5900	1
	CUST_08177	5900	1
	CUST_02864	5900	1
	CUST_03093	5900	1
	CUST_03377	5900	1
	CUST_03654	5900	1
	CUST_04067	5900	1
	CUST_04079	5900	1
	CUST_04154	5900	1
	CUST_04319	5900	1
	CUST_00998	5800	1
	CUST_01272	5800	1
	CUST_01289	5800	1
	CUST_01675	5800	1
	CUST_02036	5800	1
	CUST_03342	5800	1
	CUST_03500	5800	1

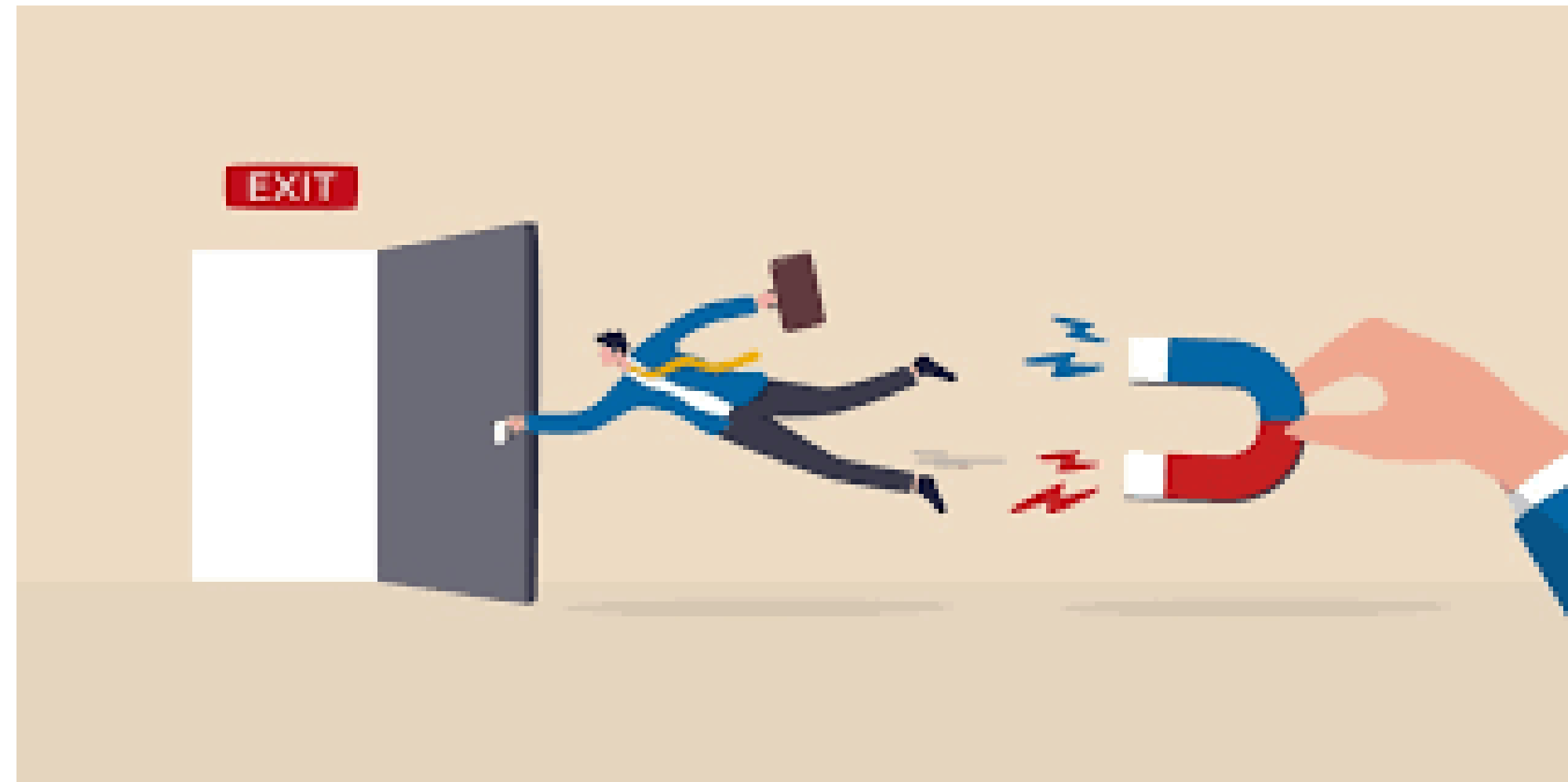
Insight

- Segments customers into value groups
- Helps prioritize high-value retention



Key Business Recommendations

- **Improve onboarding for new users**
- **Promote yearly contracts**
- **Monitor declining usage patterns**
- **Resolve billing and technical issues faster**
- **Use churn risk scores for targeted retention**
- **Offer targeted discounts before price increases**



conclusion

- Performed end-to-end customer churn analysis using SQL
- Analyzed engagement, pricing, and support data to find churn drivers
- Built churn risk indicators to identify high-risk customers
- Delivered data-driven recommendations to reduce churn



THANK
YOU